

June 2026



ENERGY STATEMENT

Demolition of Approved Barn Conversion & Construction of Replacement Dwellings & Associated Works,
PA24/07287.

Barn E at Gwarnik Mill Farm, St Allen, Truro, TR4 9QU.

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1.0 Introduction

Cornwall Planning Group is an architectural firm based in Hayle, Cornwall, with a focus on planning-led designs. Our goal is to create high-quality builds across the country, emphasising practical yet ambitious designs.

We collaborate closely with our clients to develop bespoke, contemporary designs with a strong emphasis on energy efficiency. Additionally, we have in-house Energy Assessors (OCDEAs) who specialise in completing SAPs, EPCs, and air tightness/pressure testing. Our SAP team have undertaken training to a government standard to ensure accurate SAP calculations, minimising costly redesigns, and streamlining the planning process. These services complement our building plans, enabling us to efficiently and effectively address any issues that may arise.

The purpose of this report is to help interested parties understand the energy consumption and performance of the proposed development to ensure compliance with specific regulations set out by Cornwall Council and Building Regulations.

2.0 Executive Summary

The Energy Statement, together with its associated documentation, has been prepared to substantiate a planning application for a prospective residential development. The purpose of this statement is to outline our meticulous consideration and selection of energy-efficient, low-carbon, and renewable energy measures for the aforementioned project. These options were comprehensively assessed to ascertain their compatibility with our scheme, duly weighing sustainability, environmental accountability, and feasibility.

The SAP Calculations have been used to demonstrate compliance with the planning requirement for POLICY SEC1. These calculations were prepared for the development based on the construction specifications. This information is enclosed in a separate document, which has also been uploaded for your reference.

In accordance with CEDPD Policy SEC1, the data gathered from the SAP calculations have also been input into the Energy Summary Tool SAP.

3.0 Local Planning Policy: Cornwall Council

The Climate Emergency DPD will help Cornwall Council address climate change in planning decisions. Policy SEC1 Sustainable Energy and Construction Part 2b specifically targets the energy use of new-build homes, aiming to achieve net zero operational energy demand. In essence, there are three key elements to this policy:

The three key elements are:

- A space heating energy threshold of 30 kWh/m²/year
- A total energy threshold of 40 kWh/m²/year
- A mandatory renewable energy provision equivalent to the total energy demand.

Furthermore, the policy outlines guidelines for compliance and the requisite evidence for inclusion in a planning application.

4.0 Sustainable Energy Design Strategy

4.1 Design Elements:

The energy performance of a building is affected by factors such as its design, construction, and usage. Our in-house design team has incorporated a well-insulated building structure to minimise energy consumption for environmental conditioning, thereby reducing heat loss and reliance on space heating. This Fabric-first approach aligns with passive design principles and is highly recommended in modern construction practices.

To achieve this, we will adhere to specific design and construction methods:

- Meet designated U-values using the design team's specifications.
- Install double-glazed windows throughout the building.
- Determine Thermal Bridging Values (PSI values) based on independently calculated values.
- Utilize a Mechanical Ventilation and Heat Recovery Unit (MVHR) within the building.
- Install an Air Source Heat Pump.
- Use energy-efficient lighting with a minimum power of 5W and an efficiency of 80 lm/W.
- Target an air permeability rate of 5.0 or below.

4.2 Solar Gains and Renewable Energy

The building has been designed to maximize solar gains, with strategic openings to capture sunlight during the day and evening. This has been achieved by orienting the building in a specific position.

Although the site location was unsuitable for wind and hydro technologies, and thermal technology did not provide sufficient returns to be viable, a PV array has been chosen as the most practical alternative to reduce CO2 emissions, and it has been integrated into the design.

4.3 Heating Sources

The proposed heating system will be an air source heat pump. This heating method offers a sustainable, cost-effective, and more efficient heating system. The implementation of a heat pump will effectively future proof your home against any necessary works to meet energy and carbon targets.

The building will adhere to Passive House standards, featuring high levels of insulation to minimize the need for heating throughout the year. The insulation also helps retain and release heat within the building, reducing reliance on the heating system and lowering CO2 emissions. A feasibility assessment was conducted to explore high-efficiency heating options for the development.

Water Efficiency Measures

To adhere to Policy SEC1 - Sustainable Energy and Construction from the DPD requirements, water usage must not surpass 110 litres per person per day. The specific water usage for sanitary appliances and white goods has not been determined yet, but they will be designed not to exceed the recommended 110 litres per day. As a result, for this assessment, water efficiency is considered to meet the requirements.

Overheating Mitigation

Overheating assessments will be undertaken in accordance with Approved Document O of the building regulations. This is a requirement that every new dwelling must comply with to ensure the overheating risks are effectively addressed.

5.0 Energy Efficiency

The project has undergone thermal modelling and assessment utilising SAP10 software. This procedure entails a comparative analysis of the performance of the proposed dwellings against the minimum targets stipulated by building regulation standards. The collected results have been instrumental in determining the energy target and dwelling outcomes through the utilisation of the authorised modelling software SAP 10.

The SAP assessment considers several key areas, including the DER (Dwelling Emission Rate), TER (Target Emission Rate), DFEE (Dwelling Fabric Energy Efficiency Rate), TFEE (Target Fabric Energy Efficient Rate), DPER (Dwelling Primary Energy Rate), and TPER (Target Primary Energy Rate).

The following U-Values are proposed to ensure we comply with the energy-efficient design.

	Proposed U-Values
External Wall U-Value (W/m2K)	0.14
Roof U-Value (W/m2K)	0.14
Ground Floor U-Value (W/m2K)	0.11
Window U-Value (W/m2K)	1.0
Door U-Value (W/m2K)	1.0

To ensure compliance with Cornwall Council and building regulation the requirements are as followed:

Space heating demand is to be less that 30kWh/m2/annum.

Total energy demand is to be less than 40 kWh/m2/annum.

On site renewable energy. PV will be located on the proposed roof elevations allowing for optimum sun exposure.

6.0 Energy Tool Input & Results

3 - INPUT SAP (10.2) DATA



SAP Conversion Tool V2.1

Climate Zone: 4 South West England

↓ INSERT INFORMATION HERE ↓

Results			
Space heat demand	Total energy use	Renewable generation	Renewable deficit
kWh/m ² _{TFA} /yr	kWh/m ² _{GIA} /yr	% total energy	kWh/year
Required values:			
<30	<40	100%	0
29.4	33.4	108%	0
29.7	27.1	104%	0
26.0	36.3	114%	0

Inputs - general						
Quantity	Plot Name	Bedrooms	Number of storeys	SAP Floor Area	Volume	Site Exposure
				m2	m3	
Box numbers from SAP calculation printout --->				[4]	[5]	
1	EXAMPLE - Semi Detached House	3	2	93.2	235	Normal
1	Plot - 1 Semi Detached Bungalow	3	1	134.27	448	Normal
1	Plot 2 - Semi Detached Bungalow	1	1	69.12	231	Normal

↓ INSERT INFORMATION HERE ↓

Inputs - Space Heating Demand																		
Air permeability @50Pa	Total area external elements	Total area party elements*	Ventilation system	Heat recovery	Thermal mass	Fabric heat loss	Solar gains											
m3/m2/hr	m2	m2		%	kJ/m2/K	W/K	W											
[17]	[31]	[32]+[32a]+[32b]		[23c]	[35]	[37]	[83]											
							Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	193	39	MVHR	79%	124	55.6	91	139	203	285	322	357	307	287	238	162	106	77
				0%														
1	426	5	MVHR	85%	145	92.7	271	397	543	713	785	867	758	722	629	457	313	224
				0%														
1	206	5	MVHR	85%	142	49.5	116	190	300	436	498	554	483	450	365	232	138	92
				0%														

↓ INSERT INFORMATION HERE ↓

Inputs - Total Energy Use										Inputs - Renewables	
Space heat source	Heating efficiency	Space heat source	Heating efficiency	Fraction of heat	Domestic hot water source	Water heating efficiency	Hot water storage losses	Pumps and fans energy	Lighting Efficacy	Renewable Generation	
(Primary)		(Secondary)					kWh/day	kWh/year	Lumen/Watt	kWh/year	
	[206]		[207]	[201]		[216]	[48]	[231]		[233]	
Heat pump - air to water	311%				Heat pump - air to water	189%	1.4	189	100	-3160	
Heat pump - air to water	365%				Heat pump - air to water	199%	1.99	373	80	-3554	
Heat pump - air to water	401%				Heat pump - air to water	205%	1.99	170	80	-2706	

7.0 Conclusion

In conclusion, the design has shown it will be highly efficient in meeting all requirements to satisfy Policy SEC1 Part 2b of The Climate Emergency DPD.

No offset payment is required due to meeting renewable energy targets.